

Author / Lead Developer: Spoorthi Kiran	Target Platform: Web & Progressive Web App (PWA)	GitHub Repository: github.com/spoorthikiran13/civictrack
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1. Executive Summary & Problem Statement

Municipal public works departments encounter major operational delays due to unorganized citizen complaints, lack of automated department routing, duplicate work order proliferation, and network dead zones preventing field crews from receiving or updating tickets in real-time.

CivicTrack introduces an autonomous municipal dispatch platform that analyzes text and audio complaints, routes issues directly to appropriate divisions, strictly eliminates duplicates, and supports full offline PWA caching.

2. Core Architectural Pillars

- ### A. AI Department Auto-Routing & SLA Engine

Extracts semantic keywords from voice recordings and descriptions to route tickets directly to division units (e.g., Roads & Pavement, Street Lighting Grid, Water Authority) with dedicated SLA targets (8h to 36h).
- ### B. Multi-Factor Duplicate Prevention Radar

Evaluates category affinity, Haversine GPS proximity ($\leq 150\text{m}$), municipal ward jurisdiction, and address matching. Strictly blocks redundant ticket creation and merges citizen photos/evidence into the active ticket with +1 urgency score.
- ### C. Offline PWA & Background Synchronization

Features Service Worker asset caching, IndexedDB outbox storage, and transparent online/offline network detection. Reports filed with zero network connectivity are queued securely and auto-dispatched once connection restores.

3. Municipal Department Routing Specification

CATEGORY	CODE	ASSIGNED MUNICIPAL DEPARTMENT & DIVISION	SLA
Roads & Potholes	DEPT - TRANS - 01	Dept. of Transportation (Pavement Division)	24 Hours
Street Lighting	DEPT - ELEC - 02	Bureau of Street Lighting & High-Voltage Grid	24 Hours
Water & Sewage	DEPT - WATER - 03	Municipal Water & Sewerage Authority	12 Hours
Waste & Sanitation	DEPT - WASTE - 04	Dept. of Public Sanitation & Waste Management	24 Hours
Traffic & Signals	DEPT - TRAF - 05	Traffic Operations & Intelligent Transit Systems	8 Hours
Parks & Trees	DEPT - PARK - 06	Dept. of Parks, Urban Forestry & Grounds	36 Hours

4. Automated Duplicate Detection Formula

Composite Match Score = (Category Affinity * 0.40) + (Ward Jurisdiction * 0.20)
+ (Spatial Proximity <= 150m * 0.25) + (Lexical Text Match * 0.15)

Threshold: >= 55% identifies an active duplicate. The agent disallows duplicate ticket registration.

5. Technology Stack & Deployment Architecture

Frontend Framework
React 19 & TypeScript 5

Offline Engine
Service Workers & IndexedDB Outbox

Build & Bundling
Vite 8 & ESM Architecture

Styling & UI
Tailwind CSS v4 & Lucide Icons

Spatial Mapping
Interactive GIS Ward Grid & Pins

Hosting Platforms
GitHub Pages / Cloud Run / Vercel

6. Live Presentation & Repository Links

Interactive Demo App: <https://ais-pre-t2re2hsp3i6hvlnzreoc3r-509295718433.asia-east1.run.app>

GitHub Pages Link: <https://spoorthikiran13.github.io/civictrack/>